PROJECT DOCUMENTATION

LeadPilot CRM

Customer Lead Management System

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SIC NO.	24BCSB09
ROLL NO.	4
SECTION	E2
PROJECT DURATION	3rd August 2026 – 1st September 2026
DOCUMENT VERSION	1.0

Manage Leads • Build Relationships • Grow Your Business

Table of Contents

1. Introduction
2. Problem Statement
3. Objective of the Project
4. User Requirements
5. System Architecture
6. Technology Stack
7. Core Modules & Features
8. REST API Documentation (Swagger)
9. Deployment Details
10. Project Timeline
11. Conclusion & Future Scope

1. Introduction

LeadPilot CRM is a full-stack Customer Lead Management System built to help organizations capture, track, and nurture customer leads through a single, centralized platform. The system provides sales and marketing teams with the tools to manage leads, schedule and monitor follow-ups, receive real-time reminders, and gain organization-wide visibility into pipeline performance through an interactive admin dashboard. LeadPilot CRM is designed with a modular architecture, allowing administrators to configure lead categories, subcategories, and lead sources, and to manage executives and their access with fine-grained security controls.

The application follows a three-tier architecture consisting of an Angular-based front end, a Spring Boot middleware layer exposing secure REST APIs, and a MySQL relational database for persistent storage. The system is cloud-deployed, making it accessible from anywhere without any local installation.

2. Problem Statement

Many small and mid-sized businesses continue to manage customer leads manually using spreadsheets, notebooks, or scattered messaging apps. This unstructured approach leads to several recurring problems:

- Leads collected from multiple sources (calls, WhatsApp, email, walk-ins) are not stored in one place.
- Sales executives frequently miss scheduled follow-ups due to the lack of automated reminders.
- Management has no real-time visibility into how many leads are new, active, hot, or converted.
- There is no structured way to classify leads by category, sub-category, or source for better targeting.
- Manually tracking executive performance and lead assignment becomes error-prone as the team grows.
- Sensitive customer and business data is not protected by any formal access control or security policy.

LeadPilot CRM directly addresses these problems by offering a centralized, secure, and automated platform for end-to-end lead management.

3. Objective of the Project

- To design and develop a centralized web-based CRM for managing customer leads.
- To automate follow-up scheduling and provide real-time reminder notifications.
- To provide administrators with a real-time analytical dashboard of CRM activity.
- To allow flexible configuration of lead categories, sub-categories, and lead sources.
- To implement secure, role-based access with strong authentication and account protection.
- To deploy the application on the cloud for anytime, anywhere accessibility.

4. User Requirements

4.1 Functional Requirements

- User authentication with workspace creation and secure sign-in.
- Create, view, update, and delete (CRUD) customer lead records.
- Assign leads to executives and track lead status and priority (New, Warm, Hot, Contacted, Won, Lost).
- Schedule, reschedule, complete, or cancel follow-ups linked to a specific lead.
- Receive real-time, auto-refreshing reminder notifications for due and upcoming follow-ups.
- Maintain a chronological activity log and notes for every customer lead.
- Configure and manage lead categories, sub-categories, and lead sources.
- Create, view, edit, activate, and deactivate executive (user) accounts.
- View an administrator dashboard with real-time statistics on leads, follow-ups, and executives.
- Perform global search across leads, categories, and follow-ups from a single search bar.
- Manage personal profile information and change account password securely.
- Configure system-wide general settings (CRM name, time zone, date/time format, notification behaviour).

4.2 Non-Functional Requirements

- **Security:** Password policies, login-attempt lockout protection, optional two-factor authentication, and session timeout controls.
- **Usability:** A clean, responsive, and intuitive user interface accessible on desktop and mobile browsers, with light and dark theme support.
- **Performance:** Fast page loads and near real-time updates for reminders and dashboard statistics.
- **Scalability:** A modular REST API design that supports the addition of new modules without impacting existing functionality.
- **Availability:** Cloud hosting for the front end and back end ensures 24x7 accessibility.
- **Maintainability:** Clear separation of concerns across front end, middleware, and database layers.

5. System Architecture

LeadPilot CRM follows a standard three-tier architecture that cleanly separates presentation, business logic, and data persistence:

- **Presentation Layer (Frontend):** An Angular 15 single-page application that consumes REST APIs and renders the dashboard, lead management screens, and configuration panels.
- **Application / Middleware Layer:** A Spring Boot (Java) service that exposes secure REST endpoints, implements business logic and validation, and manages authentication and authorization.
- **Data Layer:** A MySQL relational database that stores users, leads, follow-ups, reminders, categories, sources, and configuration data.

The Angular frontend and Spring Boot backend communicate over HTTPS using JSON payloads. The backend exposes its REST API contract through Swagger / OpenAPI documentation for easy testing and integration.

6. Technology Stack

Layer	Technology	Purpose
Frontend	Angular 15, TypeScript, HTML5, CSS3	Building the single-page application, dashboard UI, and dynamic forms.
Middleware / Backend	Spring Boot (Java), Spring Security, Spring Data JPA	Exposing REST APIs, handling business logic, authentication, and authorization.
Database	MySQL	Persistent relational storage for leads, users, follow-ups, and configuration data.
API Documentation	Swagger / OpenAPI	Interactive documentation and testing of REST endpoints.
Frontend Hosting	Netlify	Cloud hosting and continuous deployment of the Angular build.
Backend Hosting	Render	Cloud hosting of the Spring Boot REST API service.
Version Control	Git & GitHub	Source code management and collaboration.

6.1 Why This Stack?

- **Angular 15** offers a robust, component-based structure well suited to a data-heavy admin application with multiple modules (leads, follow-ups, settings, etc.).
- **Spring Boot** provides a mature, production-ready framework for building secure, well-structured REST APIs with built-in support for validation, security, and dependency injection.
- **MySQL** offers reliable relational storage, well suited to the structured, related nature of CRM data such as leads, executives, categories, and follow-ups.
- **Swagger/OpenAPI** keeps the API self-documenting, making it easy to test endpoints and onboard new developers.

7. Core Modules & Features

Module	Description
Authentication & Workspace	Secure sign-in, new workspace creation, and session management for administrators.
Admin Dashboard	Real-time cards for total, active, new, pending, hot, closed, and lost leads, along with recent leads, today's follow-ups, notes, and upcoming visits.

Module	Description
Customer Leads	Add, search, filter, view, and manage leads with status, priority, category, source, and assigned executive.
Lead Detail View	Tabbed view of a lead's overview, chronological activity log, scheduled follow-ups, and notes.
Follow-Up Management	Create, reschedule, complete, or cancel follow-ups, with a live summary of scheduled, completed, missed, and cancelled items.
Reminders	Real-time, auto-refreshing reminder notifications for due and upcoming follow-ups, with read/dismiss/restore actions.
Lead Configuration	Manage lead categories, sub-categories, and lead sources, including activation and deactivation.
Executive Management	Add, view, edit, activate, and deactivate executive (team member) accounts with role and department details.
Settings	General preferences (CRM name, time zone, date/time format, notification behaviour) and a dedicated security centre.
Security Centre	Password policy configuration, login-attempt lockout protection, two-factor authentication, and session controls.
Profile	View and edit personal account details and change account password.
Global Search	Unified search bar that routes directly to the relevant leads, categories, or follow-up records.

8. REST API Documentation (Swagger)

The Spring Boot backend exposes a modular set of RESTful endpoints, documented and testable through Swagger / OpenAPI. The interactive Swagger UI can be accessed at:

https://leadpilot-crm-backend.onrender.com/swagger-ui/index.html

The table below summarizes the primary REST resources exposed by the backend, grouped by module. Endpoints follow standard REST conventions; the Swagger UI above should be treated as the single source of truth for exact paths, request/response schemas, and status codes.

Module	Method	Endpoint	Description
Authentication	POST	`/api/auth/register`	Create a new workspace / administrator account
	POST	`/api/auth/login`	Authenticate a user and issue a session/token
	POST	`/api/auth/logout`	Terminate the current session
Leads	GET	`/api/leads`	Retrieve all customer leads (supports search & filters)
	POST	`/api/leads`	Create a new customer lead
	GET	`/api/leads/{id}`	Retrieve a single lead by ID
	PUT	`/api/leads/{id}`	Update an existing lead
	DELETE	`/api/leads/{id}`	Delete a lead
Follow-Ups	GET	`/api/followups`	Retrieve all follow-up records
	POST	`/api/followups`	Create a new follow-up
	PUT	`/api/followups/{id}`	Update, reschedule, or complete a follow-up
	DELETE	`/api/followups/{id}`	Cancel / delete a follow-up
Reminders	GET	`/api/reminders`	Retrieve reminders (all / unread / due / completed)
	PUT	`/api/reminders/{id}/dismiss`	Dismiss a reminder
	PUT	`/api/reminders/{id}/restore`	Restore a dismissed reminder

Module	Method	Endpoint	Description
Categories	GET / POST	`/api/categories`	Retrieve or create lead categories
	PUT / DELETE	`/api/categories/{id}`	Update or delete a category
Sub-Categories	GET / POST	`/api/subcategories`	Retrieve or create sub-categories
	PUT / DELETE	`/api/subcategories/{id}`	Update or delete a sub-category
Lead Sources	GET / POST	`/api/sources`	Retrieve or create lead sources
	PUT / DELETE	`/api/sources/{id}`	Update, activate/deactivate, or delete a source
Executives	GET / POST	`/api/executives`	Retrieve or create executive accounts
	PUT / DELETE	`/api/executives/{id}`	Update, activate/deactivate, or delete an executive
Settings	GET / PUT	`/api/settings/general`	Retrieve or update general CRM settings
	GET / PUT	`/api/settings/security`	Retrieve or update security settings
Profile	GET / PUT	`/api/profile`	Retrieve or update the logged-in user's profile
	PUT	`/api/profile/change-password`	Change the logged-in user's password

9. Deployment Details

LeadPilot CRM is fully deployed on the cloud, with the frontend and backend hosted independently and communicating over secure HTTPS REST calls.

Component	Platform	URL
Frontend (Angular 15)	Netlify	https://leadpilot-crmapp.netlify.app
Backend REST API (Spring Boot)	Render	https://leadpilot-crm-backend.onrender.com
API Documentation (Swagger UI)	Render	https://leadpilot-crm-backend.onrender.com/swagger-ui/index.html
Database	MySQL (cloud-hosted instance)	Accessed internally by the backend service; not publicly exposed.

10. Project Timeline

The project was executed over a four-week development cycle, from 3rd August 2026 to 1st September 2026, following an iterative build-and-integrate approach.

Phase	Duration	Activities
Phase 1: Planning & Design	3 Aug – 9 Aug 2026	Requirement gathering, problem statement finalization, database schema design, wireframing of Angular screens, and API contract planning.
Phase 2: Backend Development	10 Aug – 16 Aug 2026	Spring Boot project setup, MySQL entity design, authentication module, and core REST APIs for leads and executives, documented via Swagger.
Phase 3: Frontend Development	17 Aug – 23 Aug 2026	Angular 15 application setup, dashboard UI, lead management screens, and integration with backend REST APIs.
Phase 4: Feature Completion	24 Aug – 30 Aug 2026	Follow-up management, real-time reminders, lead configuration (categories, sub-categories, sources), settings, security centre, and profile modules.
Phase 5: Testing & Deployment	31 Aug – 1 Sept 2026	End-to-end testing, bug fixes, deployment of the Angular app to Netlify and the Spring Boot API to Render, and final documentation.

11. Conclusion & Future Scope

LeadPilot CRM successfully delivers a centralized, secure, and easy-to-use platform for managing customer leads from capture to conversion. By combining an Angular 15 frontend, a Spring Boot REST API middleware, and a MySQL database, the system provides a scalable foundation that can grow with an organization's needs. Real-time dashboards, automated reminders, and configurable lead classification give sales teams the visibility and structure needed to close more deals, while the security centre ensures customer and business data remains protected.

Future enhancements that could be considered include:

- Role-based dashboards tailored to individual sales executives.
- Integration with WhatsApp Business API and email providers for automated lead capture.
- Advanced analytics and conversion-funnel reporting.
- Mobile application support for on-the-go lead and follow-up management.
- Bulk lead import/export via CSV and Excel.

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